

Introduction

A meta-analysis pools effect-size estimates from multiple studies into a single summary. The two classical approaches – the fixed-effect model and the random-effects model – rest on different assumptions about what those studies represent. Choosing between them is not a technical afterthought; it is a substantive decision about the scope of inference, and it can change both the point estimate and its uncertainty.

Prerequisites

The reader should know what an effect size is (mean difference, standardised mean difference, log-odds ratio, etc.), how its standard error is computed, and how a weighted average works. Familiarity with the `meta` or `metafor` packages is helpful.

Theory

Let $\hat{\theta}_i$ be the effect-size estimate from study i with variance v_i , for $i = 1, \dots, k$.

Fixed-effect model

Assumes all studies estimate the *same* true effect θ :

$$\hat{\theta}_i = \theta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, v_i).$$

The optimal pooled estimate is the inverse-variance-weighted mean

$$\hat{\theta}_{\text{FE}} = \frac{\sum w_i \hat{\theta}_i}{\sum w_i}, \quad w_i = 1/v_i,$$

with standard error $1/\sqrt{\sum w_i}$. Scope of inference: *the set of studies included*.

Random-effects model

Assumes each study estimates its own study-specific true effect, drawn from a distribution of true effects across a population of plausible studies:

$$\hat{\theta}_i = \theta_i + \varepsilon_i, \quad \theta_i \sim \mathcal{N}(\mu, \tau^2), \quad \varepsilon_i \sim \mathcal{N}(0, v_i).$$

The weights become $w_i^* = 1/(v_i + \tau^2)$, where τ^2 is the between-study variance, estimated by DerSimonian-Laird, REML, Paule-Mandel, or Sidik-Jonkman. The pooled estimate is $\hat{\mu} = \sum w_i^* \hat{\theta}_i / \sum w_i^*$. Scope of inference: *the population of studies from which the included ones are a sample*.

When $\tau^2 = 0$ (no heterogeneity), the random-effects model reduces to the fixed-effect model. When τ^2 is large, random-effects weights are more balanced (small studies get relatively more weight), and the random-effects CI is wider than the fixed-effect CI – properly so, since it reflects between-study variability.

Assumptions

Both models assume:

- Study-level effect estimates are approximately normal with known variance.
- Studies are independent of each other.
- No publication bias or other selection distorting the set of included studies.

The fixed-effect model additionally assumes a single true effect. The random-effects model assumes a normal distribution of true effects; the Hartung-Knapp adjustment corrects CIs under small k or imprecise τ^2 estimates.

R Implementation

```
library(meta)

dat <- data.frame(
  study = paste("Study", 1:8),
  mean_t = c(11.2, 9.8, 12.5, 10.1, 13.0, 8.7, 11.6, 10.9),
  sd_t = c(2.1, 2.4, 2.0, 2.6, 1.9, 2.8, 2.2, 2.3),
  n_t = c(40, 35, 50, 30, 45, 25, 38, 42),
  mean_c = c(9.8, 9.5, 10.1, 9.9, 10.5, 9.0, 10.0, 10.2),
  sd_c = c(2.0, 2.3, 2.1, 2.5, 2.0, 2.7, 2.1, 2.2),
  n_c = c(40, 35, 50, 30, 45, 25, 38, 42)
)

m <- metacont(n.e = n_t, mean.e = mean_t, sd.e = sd_t,
              n.c = n_c, mean.c = mean_c, sd.c = sd_c,
              studlab = study, data = dat,
              sm = "MD",
              method.tau = "REML",
              method.random.ci = "HK")

summary(m)

forest(m, leftcols = c("studlab", "n.e", "mean.e", "sd.e",
                       "n.c", "mean.c", "sd.c"))
```

`metacont()` handles continuous outcomes with mean differences. Setting `method.tau = "REML"` uses REML for τ^2 ; `method.random.ci = "HK"` applies the Hartung-Knapp correction to the random-effects confidence interval.

Output & Results

Typical output:

	MD	95%-CI	%W(fixed)	%W(random)
Fixed effect model	1.210 [0.802; 1.618]		100.0	--
Random effects model	1.195 [0.567; 1.822]		--	100.0

Quantifying heterogeneity:

```
tau^2 = 0.3521; tau = 0.5934;
I^2 = 52.8% [0.0%; 78.4%]; H = 1.46 [1.00; 2.15]
```

Test of heterogeneity:

Q	d.f.	p-value
14.83	7	0.0381

The fixed-effect and random-effects point estimates are similar here (1.21 vs. 1.20), but the random-effects confidence interval is more than 50% wider, reflecting the moderate heterogeneity ($I^2 = 53\%$).

Interpretation

Report both models when I^2 is non-negligible, and emphasise the random-effects result with its Hartung-Knapp CI as the primary inference when heterogeneity is plausible on substantive grounds. Include the τ^2 estimate and the prediction interval – the latter answers “what true effect should I expect in a future, similar study?” and is often the most informative summary for a clinician.

Practical Tips

- Choose the model based on subject-matter plausibility, not on whether heterogeneity is statistically significant – Q -tests are underpowered with few studies.
- Report the prediction interval alongside the pooled estimate; it is the honest summary of what the pooled result implies for a new study.
- Use REML or Paule-Mandel for τ^2 ; DerSimonian-Laird is the historical default but understates τ^2 with few studies.
- Never pool fewer than 3 studies without explicit caveat; with 5 or fewer, consider narrative synthesis rather than meta-analysis.
- When heterogeneity is high, investigate it with subgroup analysis and meta-regression before pooling.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis